KING SALMAN INTERNATIONAL UNIVERSITY

Field of Dental medicine

Dental biomaterials I (PDD131)



Introduction to dental biomaterials I

Assoc.Prof.Dr : Enas Mahmoud Elddamony

Date : 4/ 10/2026

Contents



- Dentistry
- Departments
- Dental materials
- Dental terminology

Course description



- 3 Credit hours
- 2 hours lecture
- 2 hours practical

Assessment of students

Midterm	**10**
Quiz	**10**
Final exam	**40** **12 min to pass**
Practical exam	**30**
Semester work (assignments, researches, presentation& posters)	**10**



Course content

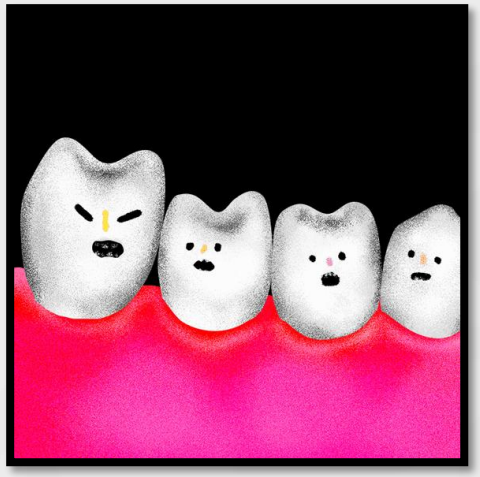
Topics	No. of lectures
Structure of matter	1
Physical properties (Thermal-Optical)	2
Mechanical properties and tests	2
Adhesion	1
Metallurgy	2
Tarnish and corrosion	1
Polymers	1

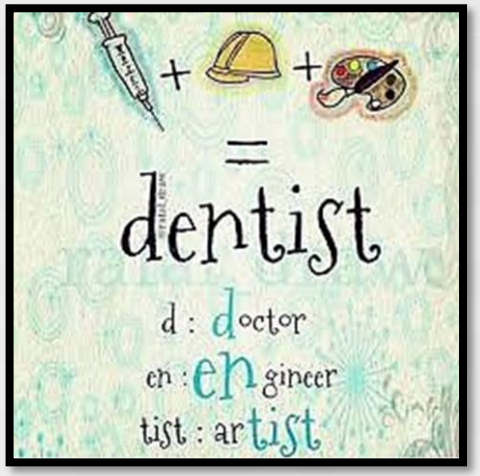


Dentistry

Science deals with:

1. Restoring esthetics
2. Relief pain
3. Restoring masticatory efficiency

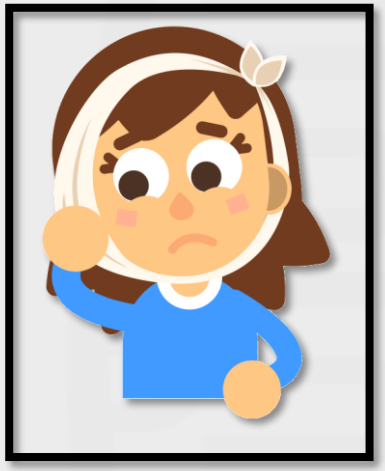


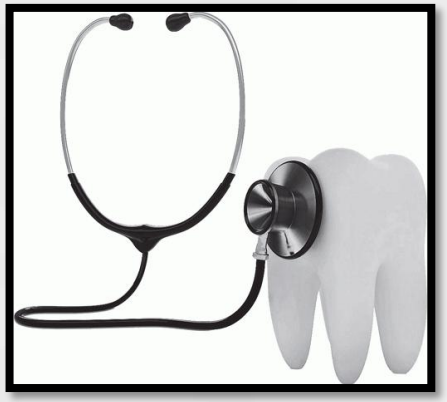












Faculty Departments

•Academic

- Oral biology
- Oral Pathology
- Oral radiology

•Clinical

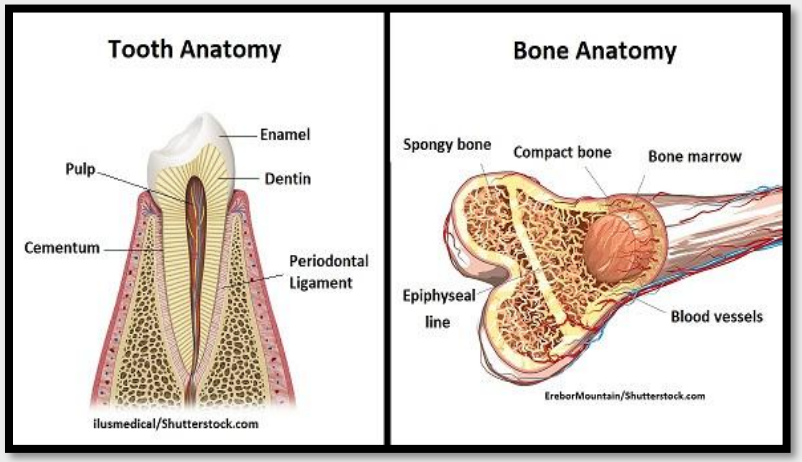
- Prosthetic dentistry
- Conservative dentistry
- Orthodontics
- Pedodontics
- Oral and maxillofacial surgery
- Oral Medicine
- Periodontology

Basic Dental Science



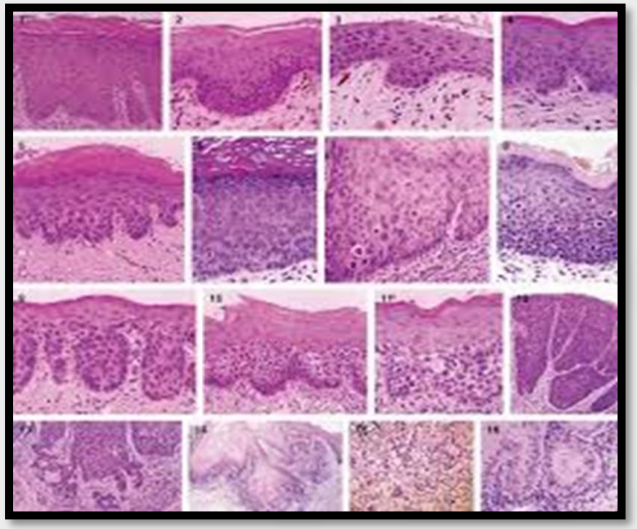
Dental anatomy
Oral biology





Oral Pathology

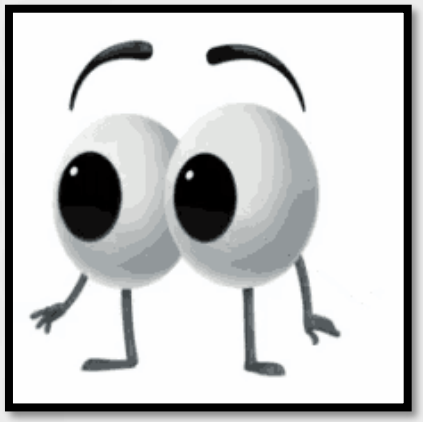




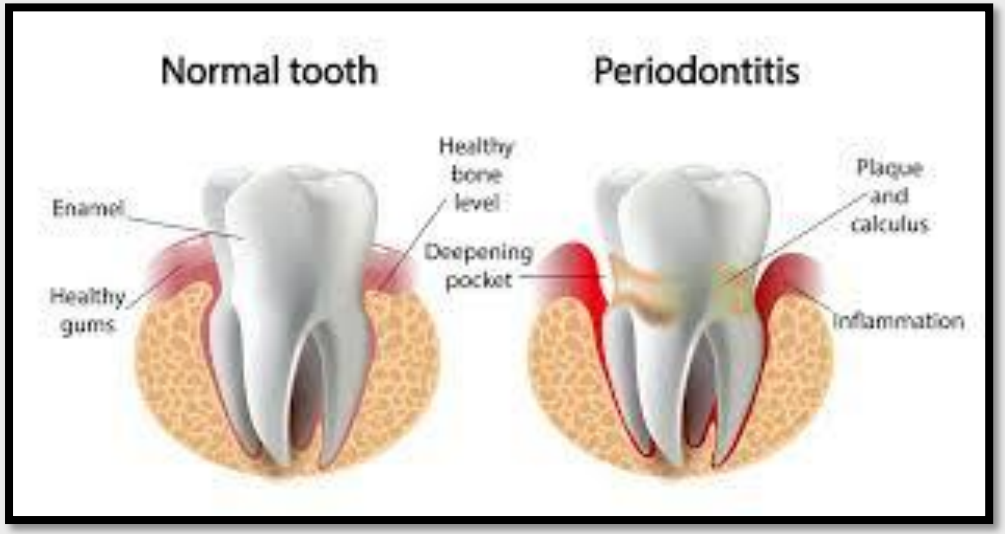
Oral Radiology



Dentists' 3rd eye



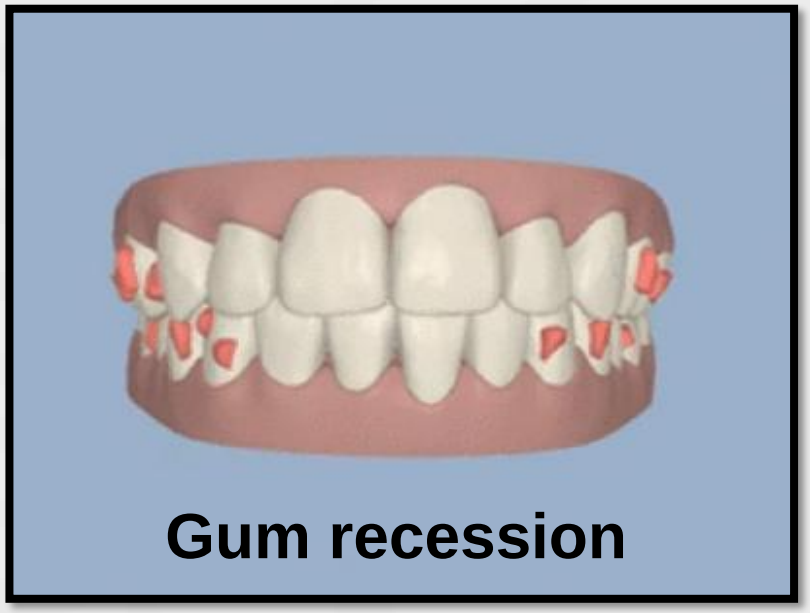
Oral medicine and periodontology





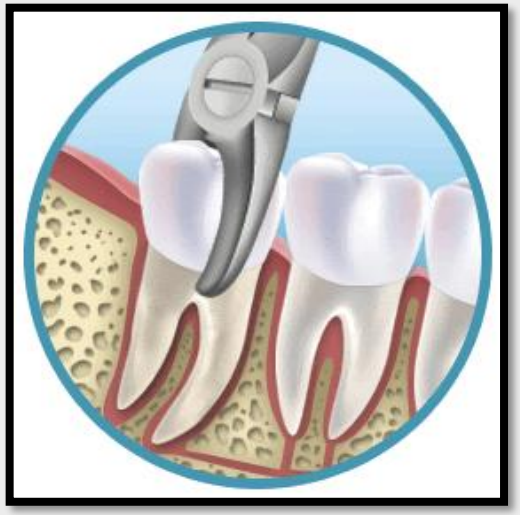


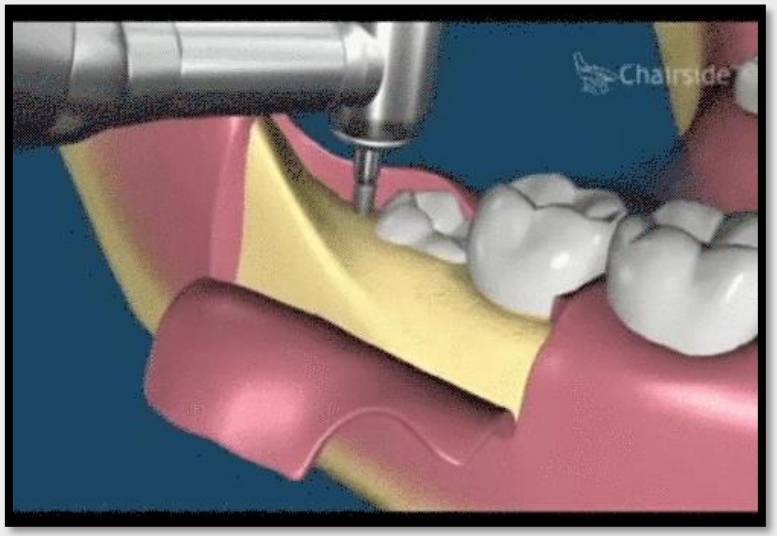
Dental plaque



Gum recession

Oral and maxillofacial surgery





Orthodontics







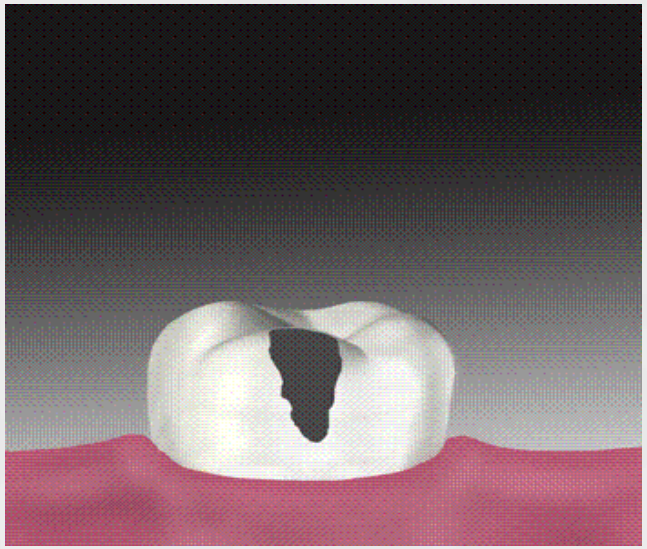
Pedodontics



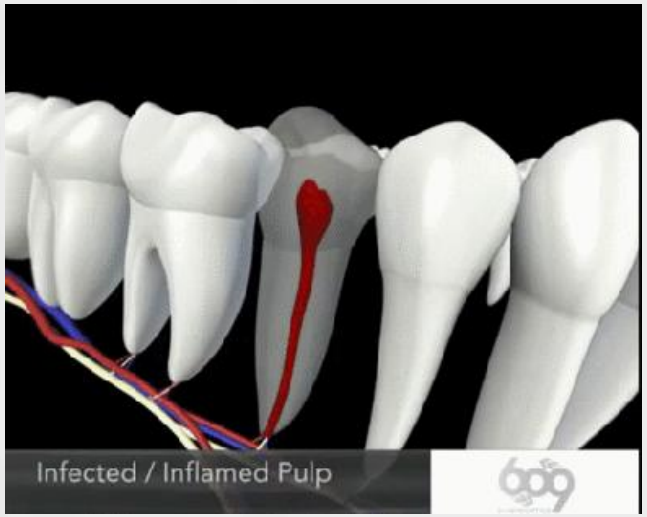




Conservative dentistry



Operative dentistry



Endodontics

Prosthetic Dentistry

- Dental Biomaterials
- Removable Prosthodontics
- Fixed prosthodontics





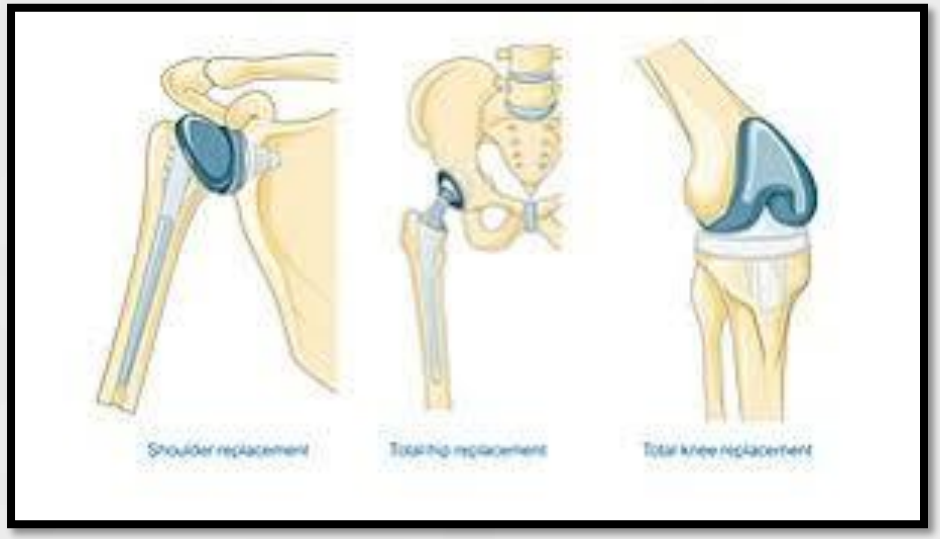




Prosthesis

- Any artificial replacement of a missing body part
- May be fixed or removable





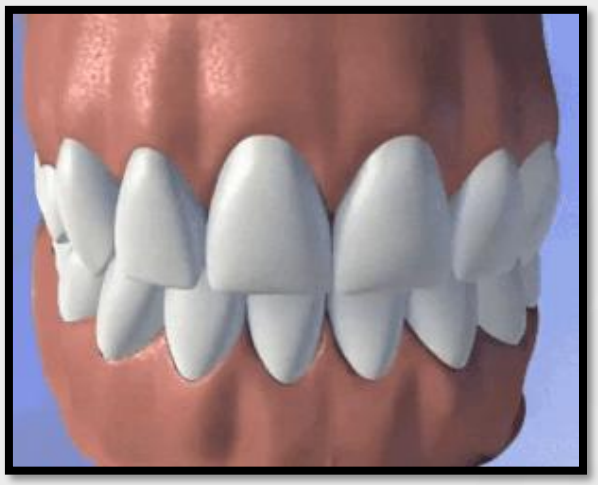
Fixed Prosthodontics

Replacement of missing tooth(teeth) by a prosthesis that is <u>not removed</u> from the mouth



Dental implants





Crown &bridge

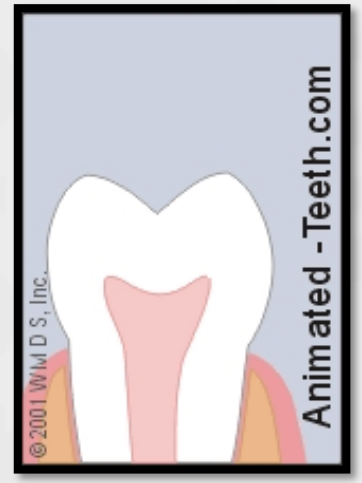




All ceramic



All metallic





Ceramo-metallic

Removable Prosthodontics

Replacement of missing tooth(teeth) by a prosthesis that is removed from the mouth





Complete denture



Partial denture

Dental Biomaterials



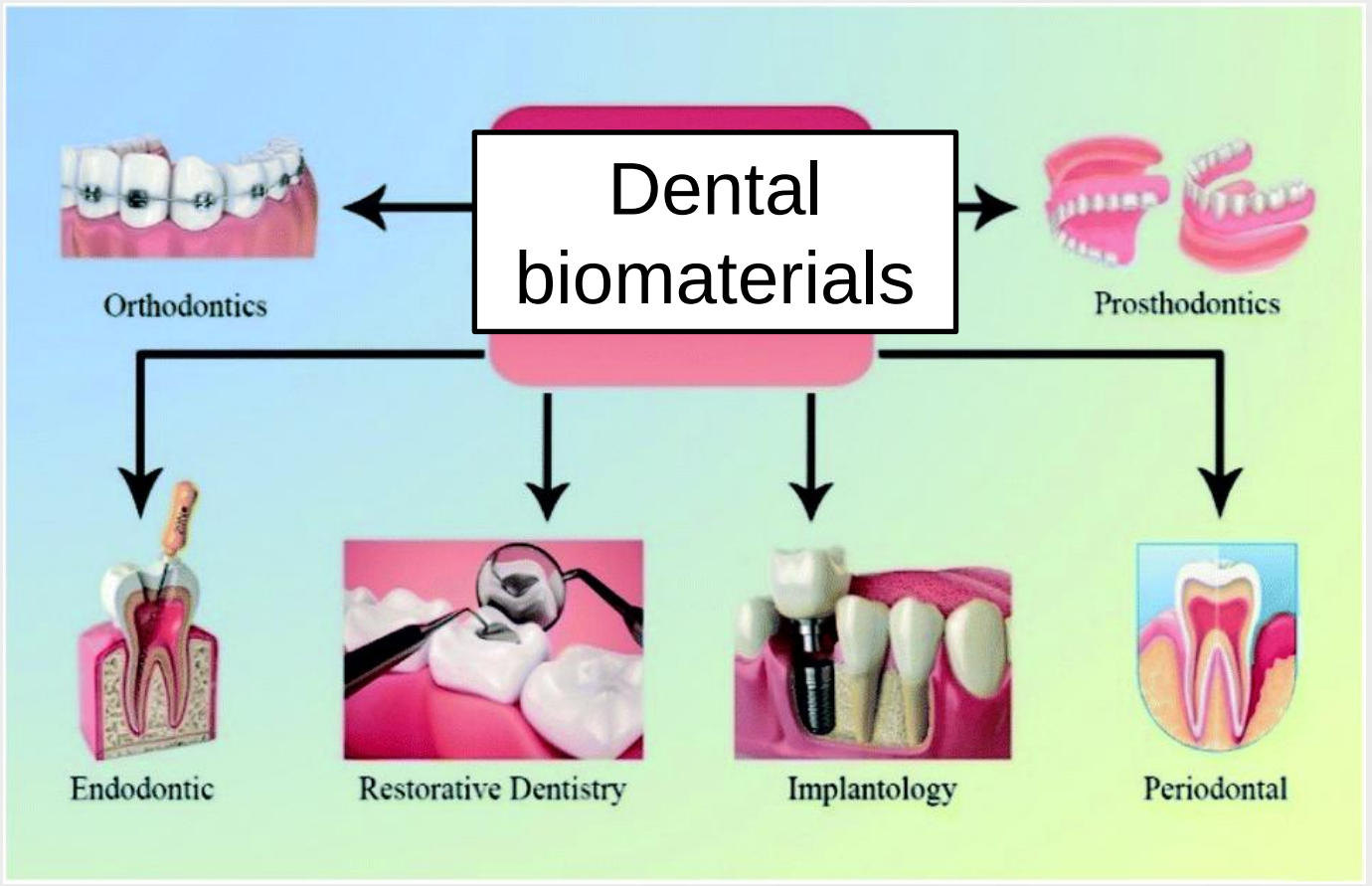




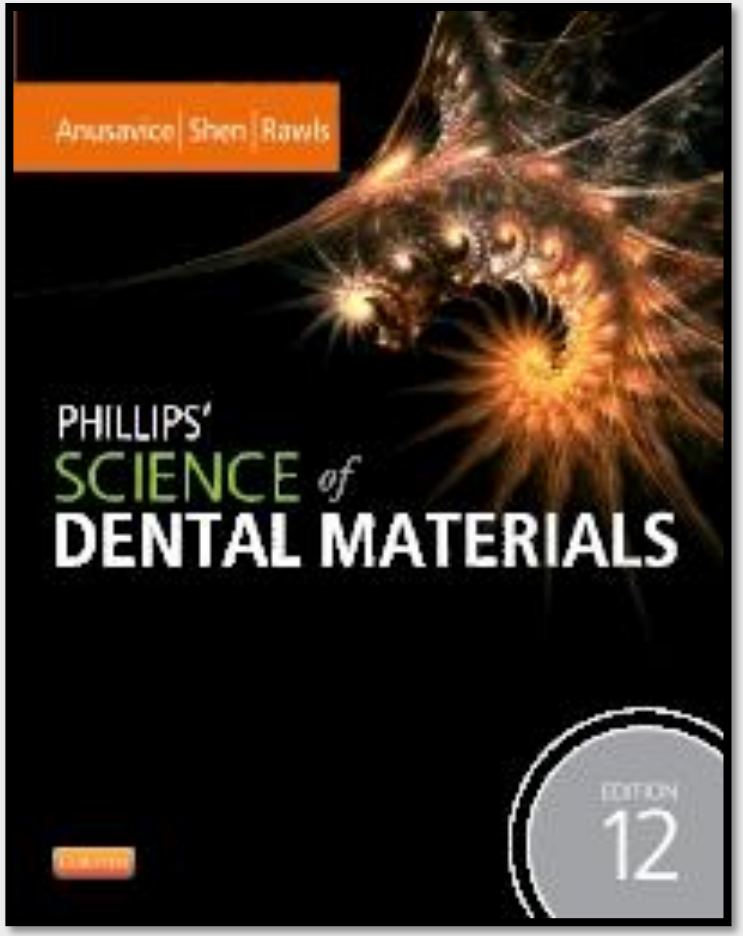


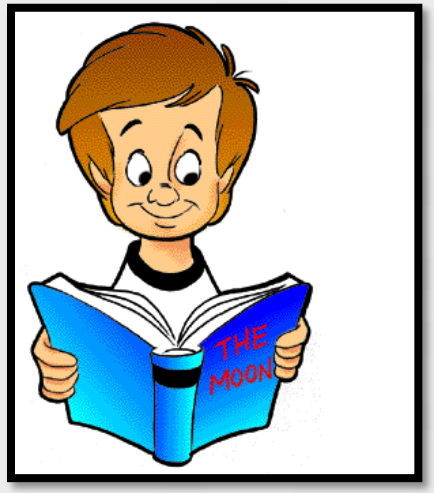


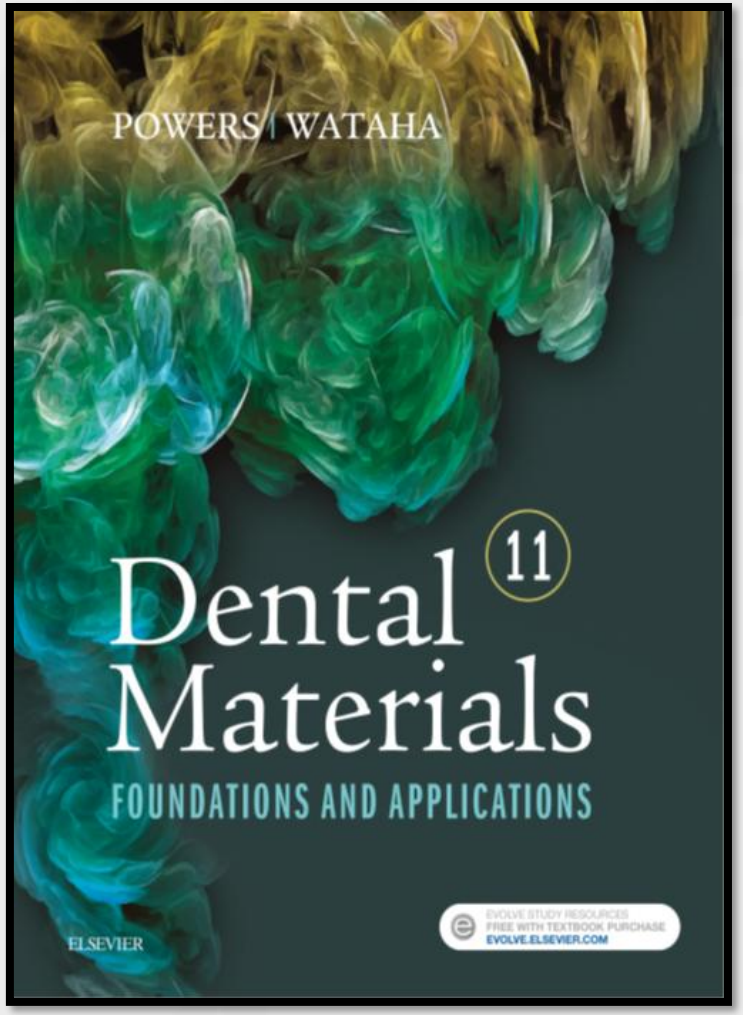
- **Biomaterials:**- synthetic materials for use in the body
- **Science of Dental biomaterials:-** science deals with materials used in oral cavity ,dental laboratories, regarding their ***composition, properties***, ***uses*** and ***manipulation***



Reference

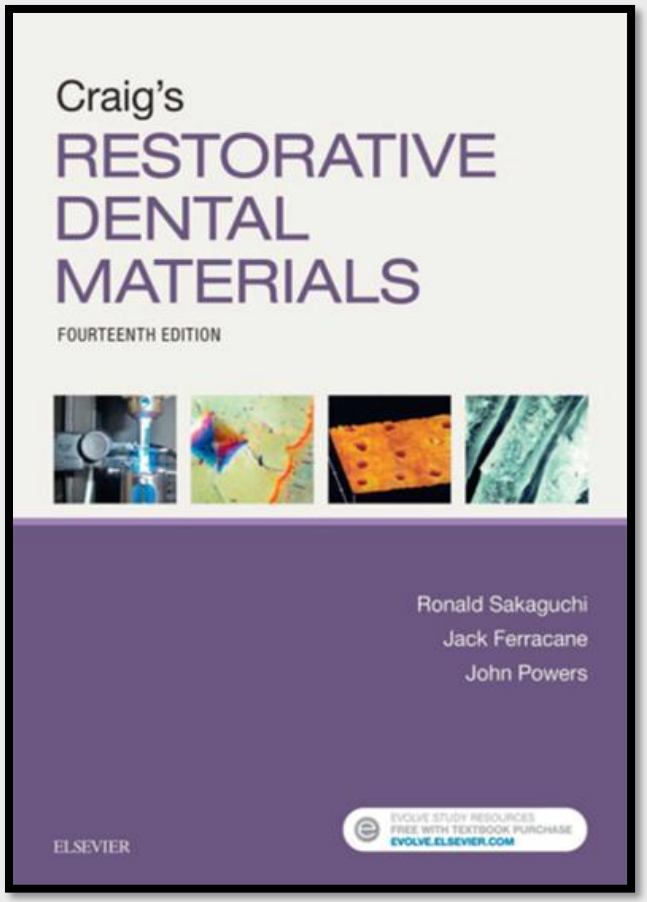

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DENTAL MATERIALS
EDITION
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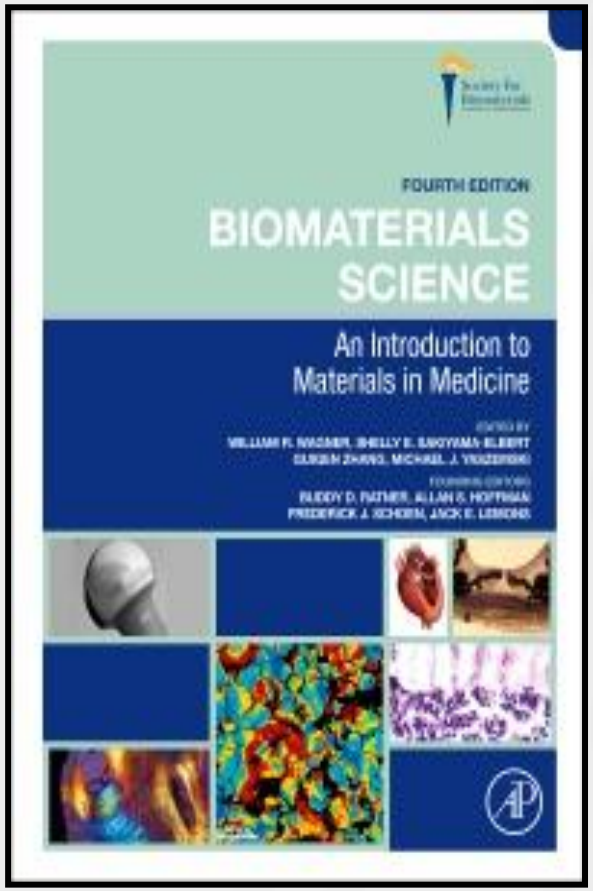

THE MOON

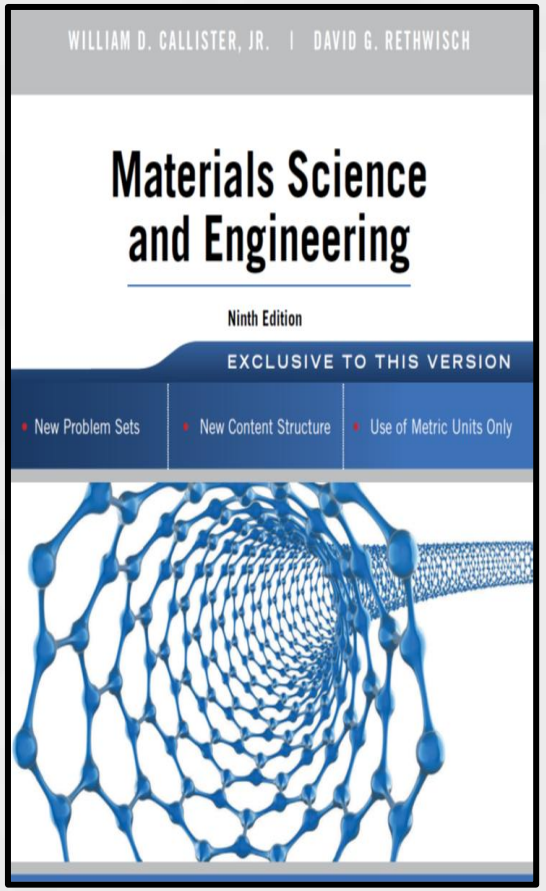

POWERS | WATAHA
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Dental Materials
FOUNDATIONS AND APPLICATIONS
ELSEVIER
EVOLVE STUDY RESOURCES
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References of Biomaterials that you may want to read :







ADA Accepted
American Dental Association ®

ANSI
American National Standards Institute



Choosing the right dental materials

Benefits of studying dental materials

- Types , indications of filling materials
- Handle materials properly
- Educate patient





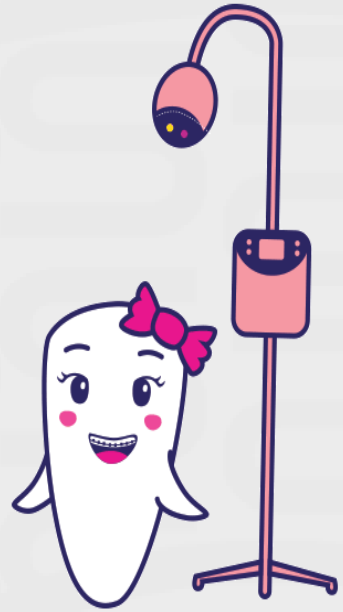


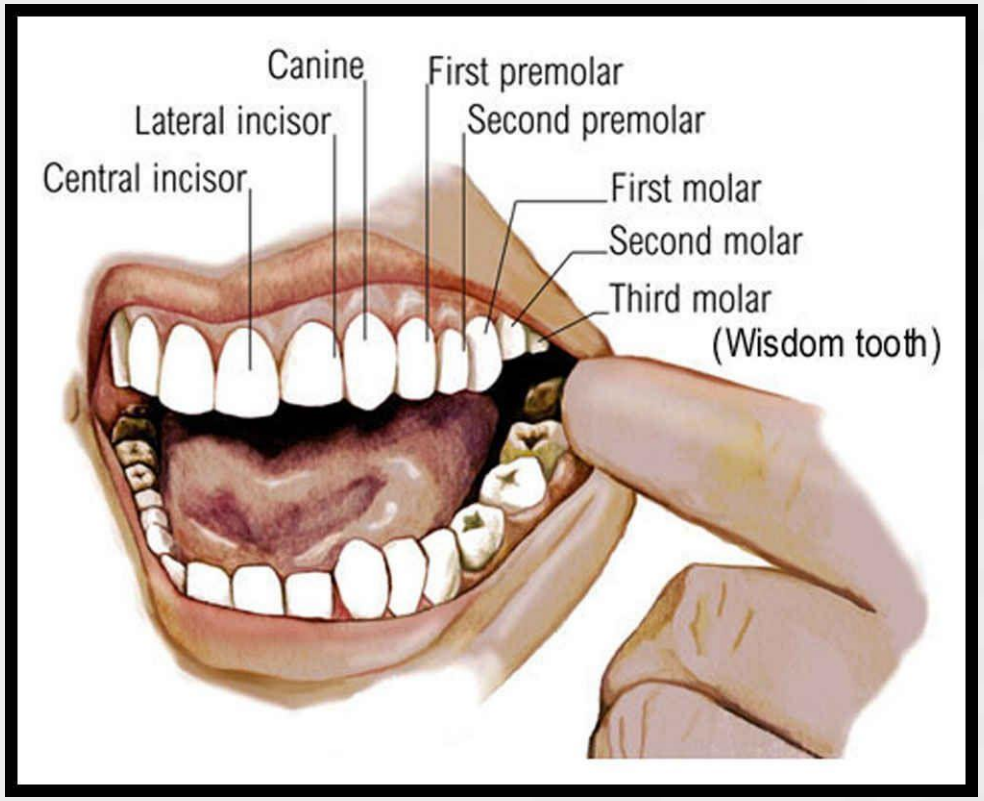




Dental terms

Anterior and posterior





Dentulous vs edentulous



Dentulous patient





Edentulous patient

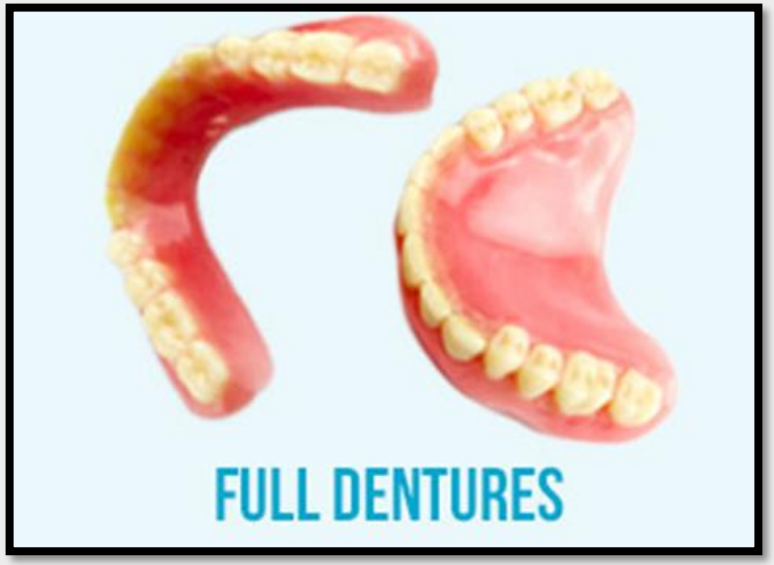






Completely Edentulous patient

Partially Edentulous patient





Denture components

- **Denture base**
- **Artificial teeth**



1. Denture base

- **Non-metallic**



Acrylic resin

- **Metallic**



Cobalt chromium alloy
Gold alloy IV

2. Artificial teeth

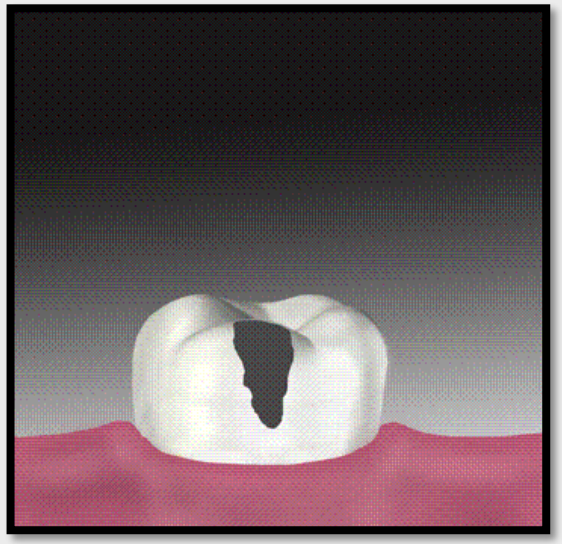
- **Acrylic resin**



- **Porcelain**



Direct and indirect restoration







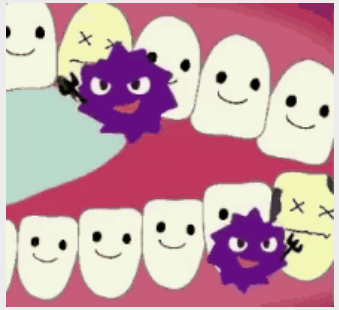


- Dental Amalgam
- Composite resin

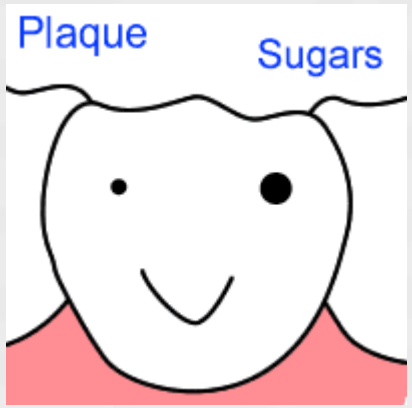
- Composite inlay
- Gold inlay
- Ceramic inlay



Dental caries







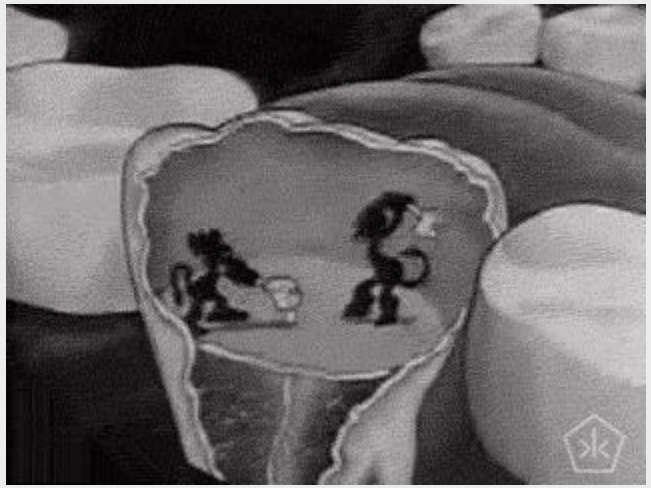
Irreversible destruction of a part of tooth

Enamel

Enamel and dentin

Reaches the pulp

Caries classes



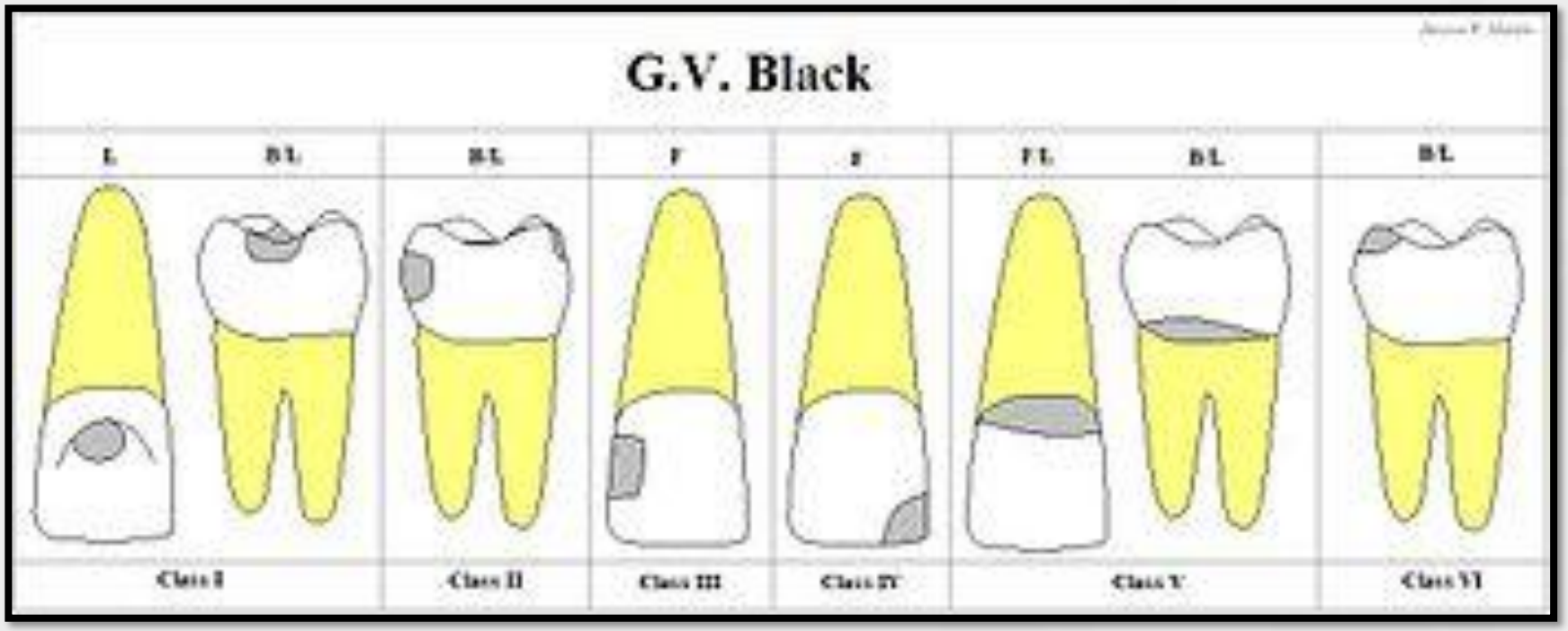
Stages	Illustration	Radiographs
Incipient		
Moderate		
Advanced		
Severe		

Classes	Illustration
Class I	
Class II	
Class III	
Class IV	
Class V	
Class VI	











Posterior teeth restoration : Strength





Class IV





Class III

Anterior teeth restoration : Esthetics



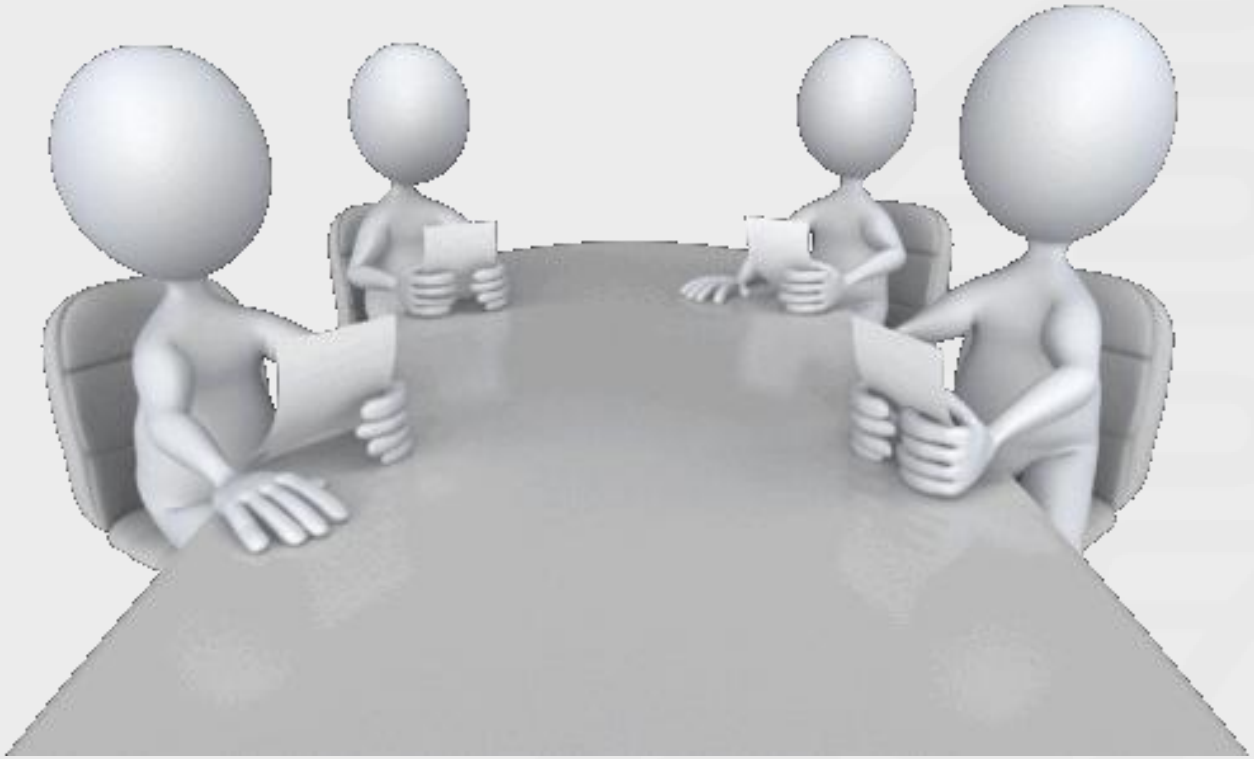






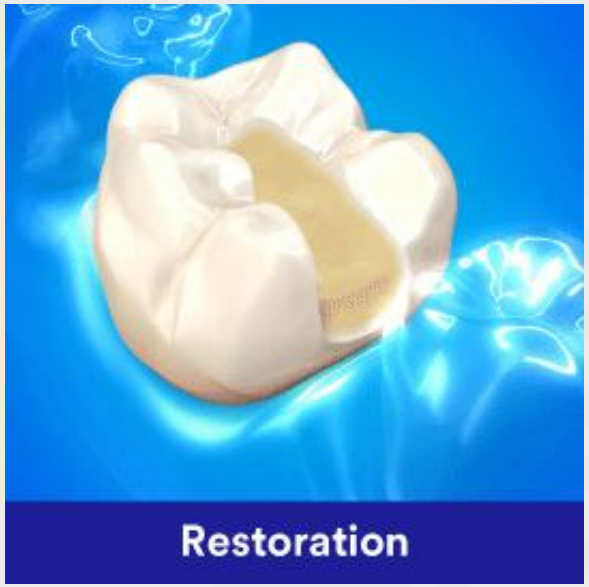


Class V




zzzzzzz







Class IV

- ✓ Composite resin
- ✓ Glass ionomer

Class II

- ✓ Dental amalgam
- ✓ Gold inlay
- ✓ Composite resin





Partially edentulous patient

Fixed prosthesis:

Bridge

Implant

Removable prosthesis:

partial denture

Complete denture

Denture base

Artificial teeth







All metallic crown

Partial denture

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THANK YOU

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